4 Random variables and their distributions

In this section we begin our systematic study of random variables. Random variables and their distributions provide a preferred language for describing and solving probabilistic questions, and should be used consistently except perhaps in elementary problems. Very soon their use will become natural.

4.1 Indicator and Bernoulli random variables

Let $(\Omega, \mathbb{P})$ be a given probability space. Recall from Definition 2.22 that a *random variable* with state space $\mathcal{X}$ is a function X from Ω to $\mathcal{X}$. In practice usually $\mathcal{X} \subset \mathbb{R}$ or $\mathcal{X} \subset \mathbb{R}^d$. Intuitively, a random variable reveals an aspect of the outcome. Also recall that X is *discrete* if its distribution is characterized by a probability mass function f_X:

$$f_X(x) = \mathbb{P}(X = x) \quad \text{and} \quad \mathbb{P}(X \in B) = \sum_{x \in B} f_X(x), \quad B \subset \mathcal{X}.$$

Since X is discrete, $f_X(x) > 0$ for at most countably many values of x.

Example 4.1 (Constant random variable)*.* Let $x_0 \in \mathcal{X}$ be fixed. Let X be a random variable such that $X(\omega) = x_0$ for all sample points ω. Then X is identically constant and we may simply write $X = x_0$. Clearly X is discrete and its probability mass function is given by

$$f_X(x) = \begin{cases} 1, & \text{if } x = x_0; \\ 0, & \text{if } x \neq x_0. \end{cases}$$

The distribution of X is called the *point mass* at x_0:

$$\mathbb{Q}_X(B) = \begin{cases} 1, & \text{if } x_0 \in B; \\ 0, & \text{if } x_0 \notin B. \end{cases}$$

Numerical constants and vectors such as 2, π and $(0, 0, 1)$ can and will be regarded as constant random variables (the last one is a constant random vector).

The simplest nonconstant random variables take two distinct values. For many purposes it is convenient to let these two values be 0 (zero) and 1 (one), so that the random variable is real-valued. In this case, the random variable is completely characterized by the event on which it is equal to 1.

Definition 4.2 (Indicator random variable)**.** *Let $A \subset \Omega$ be an event. The indicator of A is the real-valued random variable $I_A : \Omega \to \mathbb{R}$ defined by*

$$I_A(\omega) = \begin{cases} 1, & \textit{if } \omega \in A; \\ 0, & \textit{if } \omega \notin A; \end{cases} \tag{4.1}$$

By construction, we have $\{I_A = 1\} = A$ and $\{I_A = 0\} = A^c$. In words, the indicator random variable I_A tells us whether the event A occurs or not. Indicator random variables are useful mathematical tools and their uses will be illustrated throughout the course (see e.g., Section 6 where we study expectation). For now we observe that set operations of events can be naturally represented in terms of arithmetic operations regarding their indicators.

Before stating the result, we note that real-valued random variables can be combined using arithmetic operations. For example, if $X, Y : \Omega \to \mathbb{R}$ are real-valued, we may consider the following real-valued random variables:

- *sum*: $X + Y$;
- *linear combination*: $aX + bY$, where $a, b \in \mathbb{R}$;
- *product*: XY;
- *quotient*: X/Y (provided that Y never takes the value 0).

Later we will consider other transformations of random variables such as e^{tX+sY} (where $t, s \in \mathbb{R}$) and $\sqrt{X^2 + Y^2}$.

Proposition 4.3 (Algebra of indicator random variables)**.** *In the following A and B are events.*

(i) $I_\Omega = 1$ *and* $I_\emptyset = 0$. *(They are constant random variables.)*

(ii) $I_{A^c} = 1 - I_A$.

(iii) $I_{A\cap B} = I_A I_B$. *In particular,* $I_A^2 = I_A I_A = I_A$.

(iv) $I_{A\cup B} = I_A + I_B - I_{A\cap B}$. *In particular, if A and B are disjoint then* $I_{A\cup B} = I_A + I_B$.

Proof. We prove (iii) and leave the rest as exercises. (Also see Exercise 1 in Section A.4.1.)

We claim that $I_{A\cap B}(\omega) = I_A(\omega)I_B(\omega)$ for all $\omega \in \Omega$.

Case 1: $\omega \in A \cap B$. Then $I_{A\cap B}(\omega) = 1$. Since $\omega \in A$ and $\omega \in B$, we have $I_A(\omega)I_B(\omega) = 1 \cdot 1 = 1$.

Case 2: $\omega \notin A \cap B$. Then $I_{A\cap B}(\omega) = 0$. Note that $\omega \notin A$ or $\omega \notin B$ (or both). So $I_A(\omega) = 0$ or $I_B(\omega) = 0$ (or both). It follows that $I_A(\omega)I_B(\omega) = 0$. ☐

Example 4.4 (Elementary derivations involving indicators).

(i) Consider the indicator of the relative complement $A \setminus B$. Since $A \setminus B = A \cap B^c$, we have

$$I_{A\setminus B} = I_{A\cap B^c} = I_A I_{B^c} = I_A(1 - I_B) = I_A - I_A I_B = I_A - I_{A\cap B}.$$

The last expression corresponds to the representation $A = (A \cap B) \cup (A \setminus B)$ of A as a disjoint union.

(ii) We claim that $I_{A\triangle B} = |I_A - I_B|$, where the last expression is the *absolute value* of the difference between I_A and I_B. As in the proof of Proposition 4.3 we consider two cases:

Case 1: $\omega \in A\Delta B$. Then either $\omega \in A \setminus B$ or $\omega \in B \setminus A$. In both cases one of $I_A(\omega)$ and $I_B(\omega)$ is 1 and the other is 0. Thus $|I_A(\omega) - I_B(\omega)| = 1$.

Case 2: $\omega \notin A\Delta B$. Then either $\omega \in A \cap B$ (so $I_A(\omega) = I_B(\omega) = 1$) or $\omega \in (A \cup B)^c$ (so $I_A(\omega) = I_B(\omega) = 0$). Thus $|I_A(\omega) - I_B(\omega)| = 0$.

Let $X = I_A$ be an indicator random variable and consider its distribution. Let $p = \mathbb{P}(A)$ be the probability of A. It is easy to see that the probability mass function of X is given by

$$f_X(x) = \begin{cases} p, & \text{if } x = 1; \\ 1 - p, & \text{if } x = 0; \\ 0, & \text{otherwise.} \end{cases} \tag{4.2}$$

It is mathematically convenient to summarize the first two cases in (4.2) by a single equation:

$$f_X(x) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}. \tag{4.3}$$

This leads to the following basic definition:

Definition 4.5 (Bernoulli random variable)**.** *Let $p \in [0, 1]$. We say that a real-valued random variable X is a* Bernoulli(p) *random variable, written $X \sim$* Bernoulli(p)*, if it is discrete and its probability mass function of X is given by* (4.3)*. We also use* Bernoulli(p) *to denote the distribution of a* Bernoulli(p) *random variable.*[1]

Thus, if I_A is an indicator random variable, then $I_A \sim \text{Bernoulli}(p)$ where $p = \mathbb{P}(A)$. If $X \sim \text{Bernoulli}(p)$ we say that X represents a *Bernoulli trial* (a trial or a part of the experiment with two possible outcomes). In a generic discussion it is customary to say that the result is "success" if $X = 1$ and "failure" if $X = 0$. Bernoulli random variables are often used as building blocks of more complicated random variables and processes. A standard example is the construction of the binomial distribution (see Theorem 4.38).

Example 4.6 (Some transformations of Bernoulli random variables).

(i) Suppose $X_1, X_2, \ldots, X_n$ are Bernoulli random variables. For intuition, suppose $X_t = 1$ if it rains on day t. Then the *partial sum*

$$S_t = X_1 + \cdots + X_t$$

gives, for each t, the number of rainy days up to day t. In this context, it is natural to expect that the random variables $X_1, \ldots, X_n$ are related in some way (e.g. season). The stochastic dependence among the random variables is captured by the *joint distribution* of $X_1, \ldots, X_n$ (see Definition 4.20 below) which is nothing but the distribution of $(X_1, \ldots, X_n)$ regarded as an n-dimensional random vector.

(ii) Suppose $X \sim \text{Bernoulli}(p)$, and let $a, b \in \mathbb{R}$ be distinct. Let

$$Y = a + (b - a)X, \tag{4.4}$$

which is an *affine transformation* of X.[2] Note that $Y = a$ if $X = 0$ and $Y = b$ if $X = 1$. So the distribution of Y is given by $\mathbb{P}(Y = a) = p$ and

[1] Similar conventions will be adopted for other distributions such as the *binomial distribution* Binomial(n, p) in Definition 4.39.

[2] It is also common to call (4.4) a *linear* transformation of X even though strictly speaking (4.4) is not linear (in the sense of linear algebra) when $a \neq 0$.

$\mathbb{P}(Y = b) = 1 - p$. In particular, let $a = -1$ and $b = 1$. Then Y takes values in the set $\{-1, 1\}$. If furthermore $p = \frac{1}{2}$, then $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = \frac{1}{2}$. In this case we say that Y has the *Rademacher distribution*.

4.2 Distributions of univariate random variables

In this and the next subsections we state a number of important definitions which are indispensable when working with random variables. While there are quite a few definitions, it is helpful to remember that these concepts are all used to describe the distributions of random variables depending on the type of the random variable (univariate vs multivariate, discrete vs continuous) and the type of the probability (joint vs marginal, unconditional vs conditional) involved. We will become familiar with them over time.

We begin with real-valued random variables. In this case the distribution can be summarized by a single function called the *cumulative distribution function (cdf)*.

Definition 4.7 (Cumultative distribution function (cdf))**.** *Let X be a real-valued random variable and let $\mathbb{Q}_X$ be its distribution. The function $F_X : \mathbb{R} \to [0, 1]$ defined by*

$$F_X(x) = \mathbb{Q}_X((-\infty, x]) = \mathbb{P}(X \leq x) \tag{4.5}$$

is called the cumulative distribution function (cdf) of X (and of the distribution of X). Sometimes we simply call F_X the distribution function.

Example 4.8 (The cdf of Bernoulli distribution). Suppose $X \sim \text{Bernoulli}(p)$. The cdf of X is given by

$$F_X(x) = \begin{cases} 0, & \text{if } x < 0; \\ 1 - p & \text{if } 0 \leq x < 1; \\ 1, & \text{if } x \geq 1. \end{cases}$$

Since X is discrete, the cdf is *piecewise constant*. The *jumps* of F_X at $x = 0$ and $x = 1$ and their sizes correspond to the pmf of X.

Example 4.9 (The cdf of a discrete random variable). In general, if X is a discrete real-valued random variable and f_X is its pmf, then

$$F_X(x) = \sum_{y \leq x} f_X(y), \tag{4.6}$$

where the sum only involves countably many positive terms.

The next proposition summarizes the main properties of cdf.

Proposition 4.10 (Basic properties of cdf)**.** *Let F_X be the cdf of a real-valued random variable.*

(i) *For $a \leq b$ we have $F_X(b) - F_X(a) = \mathbb{Q}_X((a, b]) = \mathbb{P}(a < X \leq b)$. In particular, $F_X(x)$ is non-decreasing in x.*

(ii) $\lim_{x \to -\infty} F_X(x) = 0$ *and* $\lim_{x \to \infty} F(x) = 1$.

(iii) F_X *is right continuous. That is, for each* x *we have* $\lim_{z\downarrow x} F_X(z) = F_X(x)$*.*

(iv) If we let $F_X(x-) = \lim_{z<x, z\uparrow x} F(z)$ *be the left limit of* F_X *at* x*, then* $F_X(x) - F_X(x-) = \mathbb{Q}_X(\{x\}) = \mathbb{P}(X = x)$*.*

Proof. See Section A.4. We also note that if $F : \mathbb{R} \to [0,1]$ is a non-decreasing function satisfying properties (ii) and (iii), then F is the cdf of some real-valued random variable. □

Remark 4.11. It can be shown that the cdf (which gives $\mathbb{P}(X \le x)$ for all $x \in \mathbb{R}$) characterizes the distribution (which gives $\mathbb{P}(X \in B)$ for all $B \subset \mathbb{R}$). More precisely, if two (real-valued) random variables X and Y have the same cdfs, i.e., $F_X \equiv F_Y$, then for any $B \subset \mathbb{R}$ we have $\mathbb{Q}_X(B) = \mathbb{Q}_Y(B)$. Unfortunately, a rigorous proof of this result is beyond the scope of this course. The cdf is sometimes easier to work with since it is a function of a real variable (while the distribution is a set function).

So far we have been focusing on discrete random variables whose distributions can be characterized by probability mass functions. In many applications it is natural to consider random variables which take a *continuum* of values. These random variables are said to be *continuous.* Here is the univariate case:

Definition 4.12 (Continuous random variable (univariate)). *A real-valued random variable* X *is said to have a continuous distribution*[3] *if its distribution* $\mathbb{Q}_X$ *is given in terms of a density. That is, there exists a non-negative function* $f_X : \mathbb{R} \to [0,\infty)$ *such that for any* $B \subset \mathbb{R}$ *we have*

$$\mathbb{Q}_X(B) = \mathbb{P}(X \in B) = \int_B f_X(x)\mathrm{d}x. \tag{4.7}$$

We call f_X *the probability density function (pmf) of* X*. We often simply call* f_X *the density.*

Letting $B = [a,b]$ for $a \le b$, we have $\mathbb{P}(a \le X \le b) = \int_a^b f_X(x)\mathrm{d}x$. This gives an intuitive interpretation of the probability density function: Given $x \in \mathbb{R}$, for $\Delta x > 0$ small, we have

$$\mathbb{P}(x \le X \le x + \Delta x) \approx f_X(x)\Delta x.$$

(This holds provided that f is, say, continuous at x. See Proposition 4.17(ii).) Since $\mathbb{Q}_X(\mathbb{R}) = \mathbb{P}(X \in \mathbb{R}) = 1$, the pdf satisfies the *normalization*

$$\int_{\mathbb{R}} f_X(x)\mathrm{d}x = \int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = 1. \tag{4.8}$$

(Compare with (2.17) in the discrete case.) Conversely, any $f : \mathbb{R} \to [0,\infty)$ satisfying $\int_{\mathbb{R}} f(x)\mathrm{d}x = 1$ is the pmf of some continuous (real-valued) random variable.

We give here a basic example which generalizes Example 2.11. Several other fundamental univariate continuous distributions will be introduced in Section 5 and later in the course.

[3] Strictly speaking the distribution is said to be *absolutely continuous.* But in elementary probability the word "absolute(ly)" is often omitted.

Definition 4.13 (Uniform distribution)**.** *Let $a, b \in \mathbb{R}$ where $a < b$. A real-valued continuous random variable X is said to follow the uniform distribution on $[a, b]$, written $X \sim \text{Uniform}(a, b)$, if its density is given by*

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \textit{if } a \leq x \leq b; \\ 0 & \textit{otherwise.} \end{cases} \tag{4.9}$$

Remark 4.14 (A technical remark about the uniqueness of density)*.* Although we spoke of "the" pdf of a continuous random variable, the pdf can be slightly modified without affecting the distribution. For example, in (4.9) we may use instead the density

$$\tilde{f}_X(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b; \\ 0 & \text{otherwise.} \end{cases}$$

Note that the density is now 0 at the endpoints. Since the sets $\{a\}$ and $\{b\}$ have "zero length", for any $B \subset \mathbb{R}$ we have $\int_B f_X(x)\mathrm{d}x = \int_B \tilde{f}_X(x)\mathrm{d}x$. So $\tilde{f}$ is an equally valid density for $\text{Uniform}(a, b)$. In particular, the uniform distribution on $[a, b]$ and the uniform distribution on (a, b) are mathematically the same object.

Some general properties of continuous distributions are given in the next two propositions.

Proposition 4.15 (Continuous distributions do not charge countable sets)**.** *Let X be a continuous random variable. Then for any x we have $\mathbb{P}(X = x) = 0$. More generally, $\mathbb{P}(X \in B) = 0$ for any countable subset B of $\mathbb{R}$.*

Proof. Let f_X be the probability density of X. Then

$$\mathbb{P}(X = x) = \int_{\{x\}} f_X(z)\mathrm{d}z = \int_x^x f_X(z)\mathrm{d}z = 0$$

since the interval $[x, x]$ (domain of integration) has zero length.

Next let $B \subset \mathbb{R}$ be a countable set. Then we may write $B = \{x_1, x_2, \ldots\}$ where $x_1, x_2, \ldots$ are the distinct elements of B. So $B = \bigcup_{i \geq 1} \{x_i\}$. By the sum rule, we have

$$\mathbb{P}(X \in B) = \sum_{i \geq 1} \mathbb{P}(X = x_i) = 0.$$

□

Remark 4.16. In implementations (e.g., simulation) continuous distributions are approximated by discrete distributions. After all, real numbers cannot be stored in the computer with infinite precision and must be truncated in some way. But this practical limitation must be distinguished from theoretical properties such as the

must distinguish theoretical properties

Proposition 4.17 (Relation between cdf and pdf)**.** *Let X be a continuous real-valued random variable with cdf F_X and pdf f_X. Then:*

(i) for any $x \in \mathbb{R}$ we have

$$F_X(x) = \int_{-\infty}^{x} f_X(z)\mathrm{d}z \tag{4.10}$$

(ii) if f_X is continuous at x then

$$f_X(x) = F_X'(x) = \frac{\mathrm{d}}{\mathrm{d}z} F_X(z)\big|_{z=x}. \tag{4.11}$$

Proof. (i) Letting $B = (-\infty, x]$ in (4.7), we have

$$F_X(x) = \mathbb{Q}_X((-\infty, x]) = \int_{(-\infty,x]} f_X(z)\mathrm{d}z = \int_{-\infty}^{x} f_X(z)\mathrm{d}z.$$

(ii) From (i), F_X is given by (4.10). By the fundamental theorem of calculus, (4.11) holds whenever f_X is continuous at x. □

Remark 4.18. Here is a slightly more general version of Proposition 4.17(ii): if F_X is the cdf of a continuous random variable, then its derivative $F_X'(x)$ (which exists for all x except a "small" set) defines a pdf of X.

Example 4.19 (The cdf of uniform distribution). Let $X \sim \text{Uniform}(a, b)$. Using Proposition 4.17(i), we see by integration that

$$F_X(x) = \begin{cases} 0, & \text{if } x < a; \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b; \\ 1, & \text{if } x > b. \end{cases}$$

When $a < x < b$, F_X is differentiable at x and we have $F_X'(x) = \frac{1}{b-a}$. Similarly, we see that $F_X'(x) = 0$ when $x < a$ or $x > b$. This recovers the density of $\text{Uniform}(a, b)$. (The values at a and b are irrelevant; see Remark 4.14.) Also note that when $a = 0$ and $b = 1$ then $F_X(x) = x$ for $0 \leq x \leq 1$.

4.3 Joint, marginal and conditional distributions

Next we turn to distributions involving multiple random variables, all defined on the same probability space. For concreteness, we focus on the discrete case and defer the study of continuous random vectors to a later section. To motivate the definitions to come we give a simple example. Consider a discrete bivariate random vector $Z = (X, Y)$ whose probability mass function $f_Z(z) = f_Z(x, y)$ is given by Table 4.1:

For example, if $z = (1, 2)$, then

$$f_Z(z) = f_Z(1, 2) = \mathbb{P}(Z = (1, 2)) = \mathbb{P}(X = 1, Y = 2) = 0.1.$$

The probabilities $f_Z(z)$, for z in $\mathbb{R}^2$ or simply $\{0, 1, 3\} \times \{1, 2\}$, determine the distribution of Z as a standalone (bivariate) random variable. The distribution $\mathbb{Q}_Z$ of Z is called the *joint distribution* of X and Y. We also call $f_{X,Y}(x, y) = f_Z(x, y)$ the *joint probability mass function* of X and Y. But

	$Y=1$	$Y=2$	$f_X(x)$
$X=0$	0.1	0.2	0.3
$X=1$	0.3	0.1	0.4
$X=3$	0.2	0.1	0.3
$f_Y(y)$	0.6	0.4	

Table 4.1: Joint and marginal probability mass functions of a bivariate discrete random vector.

X and Y are themselves real-valued random variables and have their respective distributions. We call $\mathbb{Q}_X$ and $\mathbb{Q}_Y$ the *marginal distributions* of X and Y respectively; they are nothing but the distributions of X and Y regarded as standalone random variables. Using the joint pmf, we can determine the *marginal probability mass functions* f_X and f_Y. For example, we have

$$\begin{aligned} f_X(0) &= \mathbb{P}(X=0, Y=1) + \mathbb{P}(X=0, Y=2) = 0.1, \\ f_Y(2) &= \mathbb{P}(X=0, Y=2) + \mathbb{P}(X=1, Y=2) + \mathbb{P}(X=3, Y=2) = 0.4. \end{aligned} \tag{4.12}$$

If we think of $(f_{X,Y}(x,y))$ as a matrix, then f_X corresponds to the *row sums* and f_Y corresponds to the *column sums*. Now suppose we only observe $Y=2$. To us X is still random and its *conditional distribution*, i.e., $\mathbb{P}(X \in B | Y = 2)$ for $B \subset \mathbb{R}$, is given by normalizing the second column in Table 4.1 by $\mathbb{P}(Y = 2)$. For example, we have

$$f_{X|Y}(1|2) = \mathbb{P}(X=1|Y=2) = \frac{f_{X,Y}(1,2)}{f_Y(2)} = \frac{0.1}{0.4} = 0.25.$$

We call $f_{X|Y}(x|y)$ the *conditional probability mass function* of X given $Y = y$. Similarly, we can talk about the conditional probability mass function $f_{Y|X}(y|x)$ of Y given $X = x$.

With the above example in mind the following definitions and results should be quite natural.

Definition 4.20 (Joint and marginal distributions)**.** *Let $X_1, \ldots, X_n$ be random variables (in general they can have different state spaces). The joint distribution of $X_1, \ldots, X_n$ is the distribution of the random vector $(X_1, \ldots, X_n)$. For each i, the marginal distribution of X_i is the distribution of X_i as a standalone random variable.*

For notational simplicity, we focus on bivariate random vectors $Z = (X, Y)$ (or (X_1, X_2)) but the reader should have no difficulty in extending the results to three or more random variables.

Proposition 4.21 (From joint distribution to marginal distribution)**.** *Let (X, Y) be a bivariate random vector where X has state space $\mathcal{X}$ and Y has state space $\mathcal{Y}$. Let $\mathbb{Q}_{(X,Y)}$ be the distribution of (X, Y) on $\mathcal{X} \times \mathcal{Y}$, $\mathbb{Q}_X$ be the distribution of X on $\mathcal{X}$, and $\mathbb{Q}_Y$ be the distribution of Y on $\mathcal{Y}$. Then, for any $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$, we have*

$$\mathbb{Q}_X(A) = \mathbb{Q}_{X,Y}(A \times \mathcal{Y}) \quad \textit{and} \quad \mathbb{Q}_Y(B) = \mathbb{Q}_{X,Y}(\mathcal{X} \times B). \tag{4.13}$$

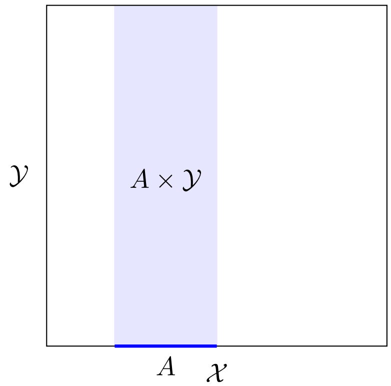

Figure 4.1: How the joint distribution determines the marginal distributions: $\mathbb{Q}_X(A) = \mathbb{Q}_{X,Y}(A \times \mathcal{Y})$ for $A \subset \mathcal{X}$. Similarly $\mathbb{Q}_Y(B) = \mathbb{Q}_{X,Y}(\mathcal{X} \times B)$ for $B \subset \mathcal{Y}$.

Proof. Consider the marginal of X. Since $\mathbb{P}(Y \in \mathcal{Y}) = 1$ by construction, we have $\mathbb{P}(X \in A) = \mathbb{P}(\{X \in A\} \cap \{Y \in \mathcal{Y}\}) = \mathbb{P}(X \in A, Y \in \mathcal{Y})$. So

$$\mathbb{Q}_X(A) = \mathbb{P}(X \in A) = \mathbb{P}(X \in A, Y \in \mathcal{Y}) = \mathbb{P}((X,Y) \in A \times \mathcal{Y}) = \mathbb{Q}_{X,Y}(A \times \mathcal{Y}).$$

Similarly, we have $\mathbb{Q}_Y(B) = \mathbb{Q}_{X,Y}(\mathcal{X} \times B)$. See Figure 4.13 for an illustration and compare with (4.12). $\square$

Remark 4.22 (Joint distribution contains more information than the marginals). Generally, we *cannot* recover the joint distribution from the marginal distributions without extra information or assumptions. This is because the marginal distributions do not tell us how the random variables are jointly related. Thus the same marginal distributions can arise from many different joint distributions. We give an example in Table 4.2.

	$Y=1$	$Y=2$	$f_X(x)$
$X=0$	0.25	0.05	0.3
$X=1$	0.1	0.3	0.4
$X=3$	0.25	0.05	0.3
$f_Y(y)$	0.6	0.4	

Table 4.2: Another joint distribution for (X,Y) whose marginal distributions are the same as those in Table 4.1.

Definition 4.23 (Joint pmf)**.** *Let $Z = (X,Y)$ be a discrete bivariate random vector. The joint pmf $f_{X,Y}(x,y)$ of X, Y is simply the pmf of Z:*

$$f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y) = \mathbb{P}((X,Y) = (x,y)) = f_Z(z), \quad z = (x,y). \tag{4.14}$$

Proposition 4.24 (From joint pmf to marginal pmf)**.** *Let (X,Y) be a discrete bivariate random vector with state space $\mathcal{X} \times \mathcal{Y}$ and joint pmf $f_{X,Y}(x,y)$. Then* *both X and Y are discrete random variables, and their marginal pmfs are given respectively by*

$$f_X(x) = \sum_{y\in\mathcal{Y}} f_{X,Y}(x,y) \quad \text{and} \quad f_Y(y) = \sum_{x\in\mathcal{X}} f_{X,Y}(x,y). \tag{4.15}$$

Note that in both sums in (4.15) *all but countably many terms are zero.*

Proof. Since (X,Y) is discrete, there exists a discrete set S, of the form $S = \{(x_1,y_1),(x_2,y_2),\dots\}$, such that $\mathbb{P}((X,Y)\in S) = 1$. It follows that $\mathbb{P}(X\in\{x_1,x_2,\dots\}) = 1$ and $\mathbb{P}(Y\in\{y_1,y_2,\dots\}) = 1$, so both X and Y are discrete.

To prove the first identity in (4.15), let $A = \{x\}$ in Proposition 4.21. Then[4]

$$\begin{aligned} f_X(x) &= \mathbb{Q}_X(x) = \mathbb{Q}_{(X,Y)}(\{x\}\times\mathcal{Y}) \\ &= \sum_{y\in\mathcal{Y}} \mathbb{Q}_{(X,Y)}(\{(x,y)\}) = \sum_{y\in\mathcal{Y}} f_{X,Y}(x,y). \end{aligned}$$

The proof of the second identity is the same. □

Definition 4.25 (Conditional distribution (discrete case))**.** *Let (X,Y) be a discrete bivariate random vector with state space $\mathcal{X}\times\mathcal{Y}$. If $x\in\mathcal{X}$ is a value for which $\mathbb{P}(X=x)>0$, the conditional distribution of Y given $X=x$ is the set function $\mathbb{P}(Y\in B|X=x)$ defined for $B\subset\mathcal{Y}$. It is described by the conditional probability mass function $f_{Y|X}(y|x)$ defined by*

$$f_{Y|X}(y|x) := \mathbb{P}(Y=y|X=x) = \frac{f_{X,Y}(x,y)}{f_Y(y)}. \tag{4.16}$$

In (4.16), the first equality holds by definition (emphasized in this particular case by the symbol :=), and the second equality follows from the definition of conditional probability:

$$\mathbb{P}(Y=y|X=x) = \frac{\mathbb{P}(X=x,Y=y)}{\mathbb{P}(Y=y)} = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

Clearly the conditional pmf satisfies the normalization $\sum_{y\in\mathcal{Y}} f_{Y|X}(y|x) = 1$. An important consequence of (4.16) is the following *product rule*:

$$f_{X,Y}(x,y) = f_X(x) f_{Y|X}(y|x). \tag{4.17}$$

So to specify the joint distribution of a (discrete) bivariate random vector, we may first specify the marginal distribution of one variable and then specify the conditional distribution of the second variable given the first one. This formula can be easily extended to any number of random variables. For example, using

[4]The third equality below make sense since we may replace $\mathcal{Y}$ by the discrete set $\{y\in\mathcal{Y} : f_Y(y)>0\}$ without affecting the probability.

self-explanatory notations, for a (discrete) random vector $(X_1, \ldots, X_n)$ with n components, the joint pmf admits the following decomposition:

$$\begin{aligned} &f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) \\ &= f_{X_1}(x_1) f_{X_2|X_1}(x_2|x_1) f_{X_3|X_1,X_2}(x_3|x_1,x_2)\cdots f_{X_n|X_1,\ldots,X_{n-1}}(x_n|x_1,\ldots,x_{n-1}). \end{aligned} \tag{4.18}$$

This decomposition is useful for describing a random process consisting of several (time, physical or conceptual) steps. From (4.18), we see that a joint distribution is generally a rather complicated object since it encodes how the conditional distributions depend on the other variables. *A key idea in probability theory is to impose, or exploit, special structures which simplify the analysis of (certain aspects of) the joint distribution.* One such structure is *independence* which will be explained in Section 4.4.

Example 4.26. Consider the random variables X and Y given in Exercise 3 in Section A.3.1. The marginal pmf of X is $f_X(x) = \frac{1}{100}$, $x \in [100]$[5] and the conditional pmf of Y given X is

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{x}, & \text{if } y \in [x]; \\ 0, & \text{otherwise.} \end{cases}$$

By (4.17), the joint pmf of (X, Y) is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{100x}, & \text{if } x \in [100] \text{ and } y \in [x]; \\ 0, & \text{otherwise.} \end{cases}$$

We also compute the marginal pmf of Y. From (4.15), for $y \in [100]$ we have

$$f_Y(y) = \sum_{x=1}^{100} f_{X,Y}(x,y) = \sum_{x=y}^{100} f_{X,Y}(x,y) = \frac{1}{100} \sum_{x=y}^{100} \frac{1}{x}.$$

Note that the second equality holds since $f_{X,Y}(x,y) = 0$ when $x < y$. Also see Exercise 3 in Section A.4.1.

Example 4.27 (Sampling without replacement). Suppose an urn contains n balls (numbered from 1 to n) and $k \leq n$ balls are drawn randomly without replacement. (A special case is worked out in Exercise 2 in Section A.3.1.) Let X_i be the i-th ball drawn, $1 \leq i \leq k$. If in each draw the remaining balls are chosen uniformly at random, the conditional pmf of X_i given $X_1, \ldots, X_{i-1}$ is given for distinct $x_1, \ldots, x_{i-1} \in [n]$ and $x_i \in \{1, \ldots, n\}$ by

$$p_{X_i|X_1,\ldots,X_{i-1}}(x_i|x_1,\ldots,x_{i-1}) = \begin{cases} \frac{1}{n-(i-1)}, & \text{if } x_1,\ldots,x_i \text{ are distinct;} \\ 0, & \text{otherwise.} \end{cases}$$

Sampling *with* replacement is treated in Example 4.36 below.

[5] Recall the notation $[n] = \{1, \ldots, n\}$ introduced in Section 3.3.

Remark 4.28 (Prelude to the continuous case). The concept of conditional distribution is more subtle in the (jointly) continuous case. This is because when X is continuous $\mathbb{P}(X = x) = 0$ for all x, so the conditional probability $\mathbb{P}(Y \in B|X = x)$ is *undefined* according to Definition 2.12. This conceptual difficulty will be resolved later. Nevertheless, the product rules (4.17) and (4.18) continue to hold if the (conditional) probability mass functions are replaced by (conditional) probability *density* functions.

4.4 Independence

In Definition 2.17 we defined the notion of independence between two events. Now we use this concept to define independence between random variables.

Definition 4.29 (Independence of two random variables). *Let X and Y be random variables with state spaces $\mathcal{X}$ and $\mathcal{Y}$ respectively. We say that X and Y are independent, written $X \perp\!\!\!\perp Y$ (or $X \perp\!\!\!\perp_{\mathbb{P}} Y$ if we want to stress the underlying probability measure $\mathbb{P}$), if the events $\{X \in A\}$ and $\{Y \in B\}$ are independent (in the sense of Definition 2.17) for any $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$:*

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B), \quad A \subset \mathcal{X}, B \subset \mathcal{Y}. \tag{4.19}$$

Mathematically, (4.19) states that the joint distribution (Definition 4.20) of (X, Y) is the *product* of the marginal distributions of X and Y:

$$\mathbb{Q}_{X,Y}(A \times B) = \mathbb{Q}_X(A)\mathbb{Q}_Y(B), \quad A \subset \mathcal{X}, B \subset \mathcal{Y}.$$

We see from (4.19) (and (2.12)) that

$$\mathbb{P}(X \in A|Y \in B) = \mathbb{P}(X \in A) \quad \text{and} \quad \mathbb{P}(Y \in B|X \in A) = \mathbb{P}(Y \in B)$$

whenever the conditional probability is defined. This gives the probabilistic interpretation of independence: if X and Y are independent, information from X does not affect our inference of Y and vice versa.

Recall in Exercise 6 of Section A.2.1 we showed that

$$A \perp\!\!\!\perp B \Leftrightarrow A^c \perp\!\!\!\perp B \Leftrightarrow A \perp\!\!\!\perp B^c \Leftrightarrow A^c \perp\!\!\!\perp B^c. \tag{4.20}$$

Thus the condition $A \perp\!\!\!\perp B$ is equivalent to the apparently stronger statement that "any event about A" is independent from "any event about B". The following result expresses this idea elegantly using the independence between random variables.

Proposition 4.30. *Let A and B be events. Then the indicator random variables I_A and I_B are independent (according to Definition 4.29) if and only if A and B are independent (according to Definition 2.17).*

Proof. ($\Rightarrow$) Suppose $I_A \perp\!\!\!\perp I_B$. Using (4.19), we have

$$\mathbb{P}(A \cap B) = \mathbb{P}(I_A = 1, I_B = 1) = \mathbb{P}(I_A = 1)\mathbb{P}(I_B = 1) = \mathbb{P}(A)\mathbb{P}(B).$$

So $I_A \perp\!\!\!\perp I_B$.

($\Leftarrow$) Suppose $A \perp\!\!\!\perp B$. For $C, D \subset \mathbb{R}$, consider the events $\{I_A \in C\}$ and $\{I_B \in D\}$. A moment's thought shows that $\{I_A \in C\}$ must be one of $\emptyset, A, A^c, \Omega$, and $\{I_B \in D\}$ must be one of $\emptyset, B, B^c, \Omega$. Using (4.20) and Lemma 4.31, we see that $\{I_A \in C\}$ and $\{I_B \in D\}$ are independent in all cases (Exercise: Give the details). Thus $I_A \perp\!\!\!\perp I_B$. $\square$

Lemma 4.31. *Let A and B be events. If $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$, then A and B are independent.*

Proof. Exercise. $\square$

Now we generalize to multiple random variables (and events). When $n = 2$, the following definition reduces to Definition 4.29.

Definition 4.32 (Independence of multiple random variables and events)**.** *Random variables $X_1, X_2, \ldots, X_n$ (with arbitrary state spaces $\mathcal{X}_1, \ldots, \mathcal{X}_n$) are jointly independent if their joint distribution is the product of the marginals, i.e., for $B_1 \subset \mathcal{X}_1, \ldots, B_n \subset \mathcal{X}_n$, we have*

$$\mathbb{P}(X_1 \in B_1, \ldots, X_n \in B_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i). \tag{4.21}$$

Events $A_1, \ldots, A_n$ are said to be jointly independent if their indicators $I_{A_1}, \ldots, I_{A_n}$ are jointly independent. In both cases we sometimes omit the word "jointly".

It is useful to think of (4.21) as an extension of the product rule (2.9) to multiple events. The probabilistic intuition of joint independence is the following: Suppose $X_1, \ldots, X_n$ are jointly independent. Then for any i, if A is an event expressed in terms of X_i and B is an event expressed in terms of $(X_j)_{j \neq i}$, then A and B are independent. The proof, which is more appropriate for a more advanced course, is omitted.

Example 4.33. Consider the events A, B and C in Example 2.19. Let us verify that they are not jointly independent, i.e., I_A, I_B and I_C are not jointly independent. Observe that

$$\{I_A = 1, I_B = 1, I_C = 1\} = A \cap B \cap C = \emptyset.$$

So $\mathbb{P}(A \cap B \cap C) = 0$. On the other hand, we have

$$\mathbb{P}(I_A = 1)\mathbb{P}(I_B = 1)\mathbb{P}(I_C = 1) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Thus (4.21) is violated and the events are not jointly independent. Also see Exercise 6 in Section A.4.1.

Theorem 4.34 (Joint pmf as product of marginal pmfs under independence)**.** *Let $(X_1, \ldots, X_n)$ be a discrete random vector. Then $X_1, \ldots, X_n$ are jointly independent if and only if their joint probability mass function is the product of the marginal probability mass functions: for any $x_1, \ldots, x_n$ we have*

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i). \tag{4.22}$$

Proof. ($\Rightarrow$) Suppose $X_1, \dots, X_n$ are jointly independent. Given $x_1, \dots, x_n$, let, in (4.21), $B_i = \{x_i\}$. Then the left hand side of (4.21) is $f_{X_1,\dots,X_n}(x_1, \dots, x_n)$ and the right hand side is $\prod_{i=1}^n f_{X_i}(x_i)$. Thus (4.22) holds.

($\Leftarrow$) Suppose the decomposition (4.22) holds. Let $B_1 \subset \mathcal{X}_1, \dots, B_n \subset \mathcal{X}_n$ be given. Since $(X_1, \dots, X_n)$ is discrete, we may without loss of generality replace each $\mathcal{X}_i$ by a countable set. By the sum rule, we have

$$
\begin{aligned}
\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) &= \sum_{x_1 \in B_1, \dots, x_n \in B_n} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) \\
&= \sum_{x_1 \in B_1, \dots, x_n \in B_n} \prod_{i=1}^n f_{X_i}(x_i) \\
&= \prod_{i=1}^n \sum_{x_i \in B_i} f_{X_i}(x_i) \\
&= \prod_{i=1}^n \mathbb{P}(X_i \in B_i).
\end{aligned}
$$

Thus $X_1, \dots, X_n$ are jointly independent. □

Comparing (4.22) with (4.18), we see that joint independence simplifies tremendously the structure of the joint distribution.

The following definition is fundamental because it provides a precise mathematical model for "repeating an experiment under the same condition".

Definition 4.35 (Independent and identically distributed (i.i.d.))**.** *Let $X_1, \dots, X_n$ be random variables with a common state space $\mathcal{X}$. We say that $X_1, \dots, X_n$ are independent and identically distributed (i.i.d.) if $X_1, \dots, X_n$ have the same distribution (i.e., the set function $\mathbb{Q}_{X_i}(B) = \mathbb{P}(X_i \in B)$ is the same for all i) and are jointly independent.*

Thus each X_i can be regarded as the result of the i-th (independent) trial of some experiment, and the vector $(X_1, \dots, X_n)$ gives the results over n trials. If the common distribution, say $\mathbb{Q}$, is given, we write $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathbb{Q}$ to mean that $X_1, \dots, X_n$ are independent and each X_i is distributed as $\mathbb{Q}$. For example, $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ means that $X_1, \dots, X_n$ is an i.i.d. sequence of Bernoulli(p) random variables.

Example 4.36 (Sampling with replacement). Consider the urn in Example 4.36 but now suppose that the k balls $X_1, \dots, X_k$ are drawn randomly *with* replacement. Now the conditional pmf of X_i given $X_1, \dots, X_{i-1}$ is given for *any* $x_1, \dots, x_{i-1}, x_i \in \{1, \dots, n\}$ by

$$
f_{X_i|X_1,\dots,X_{i-1}}(x_i|x_1, \dots, x_{i-1}) = \frac{1}{n}.
$$

Note that the conditional pmf does *not* depend on $x_1, \dots, x_{i-1}$: regardless of this history, each ball is drawn with the same probability $\frac{1}{n}$. This implies that X_i is independent from $X_1, \dots, X_{i-1}$ and so its marginal pmf is

$$
f_{X_i}(x_i) = \frac{1}{n}.
$$

The proof of this claim is given in Exercise 5 in Section A.4.1. It follows from (4.18) that for $(x_1, \dots, x_k) \in \{1, \dots, n\}^k$ we have

$$f_{X_1,\dots,X_k}(x_1, \dots, x_k) = \frac{1}{n^k} = \prod_{i=1}^{k} f_{X_i}(x_i).$$

So $X_1, \dots, X_k$ are i.i.d. and each X_i is uniformly distributed in $\{1, \dots, n\}$.

Remark 4.37. (Skip on first reading.) In what follows we will work with random variables with given distributions. We assume implicitly that in a given context, an appropriate probability space, usually denoted by $(\Omega, \mathbb{P})$, has been given and the required random variables can be defined on $(\Omega, \mathbb{P})$.[6] Thus generally we use different probability spaces in different results. Since our results are generally about the distributions of random variables (given the distributions of other given random variables), they do *not* depend on the choice of the underlying probability space. This expands on the discussion in Example 3.2 and is the main idea behind the opening paragraph of Section 4: random variables and their distributions provide a preferred language for describing and solving probabilistic questions.

In probability theory one frequently uses independent (even i.i.d.) random variables to construct further variables and processes. To illustrate the idea, we derive the *binomial distribution* using an i.i.d. sequence of Bernoulli random variables. More examples will be provided in Section 5.

Theorem 4.38 (Sum of i.i.d. Bernoulli random variables)**.** *Let $n \geq 1$ be an integer and $p \in [0, 1]$. Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $S = X_1 + \cdots + X_n$. Then the probability mass function of S is given by*

$$\mathbb{P}(S = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x \in \{0, 1, \dots, n\}, \tag{4.23}$$

and $\mathbb{P}(S = x) = 0$ for other values of x.

Proof. By Theorem 4.34 and (4.3), the joint pmf of $X_1, \dots, X_n$ is given by

$$f_{X_1,\dots,X_n}(x_1, \dots, x_n) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i},$$

for $(x_1, \dots, x_n) \in \{0, 1\}^n$. Note that the probability depends on $(x_1, \dots, x_n)$ via the sum $x_1 + \cdots + x_n$. That is, two sequences $(x_1, \dots, x_n)$ and $(x'_1, \dots, x'_n)$ have the same probability whenever $x_1 + \cdots + x_n = x'_1 + \cdots + x'_n$.

Since $S_n = X_1 + \cdots + X_n$ and $\mathbb{P}(X_i \in \{0, 1\}) = 1$, we have $\mathbb{P}(S_n \in$

[6] The construction of appropriate probability spaces, especially when continuous or infinitely many random variables are involved, is beyond the scope of this course.

$\{0, 1, \ldots, n\}) = 1$. Let $x \in \{0, 1, \ldots, n\}$. By the sum rule, we have

$$\begin{aligned}
\mathbb{P}(S = x) &= \sum_{(x_1,\ldots,x_n)\in\{0,1\}^n : \sum_i x_i = x} f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) \\
&= \sum_{(x_1,\ldots,x_n)\in\{0,1\}^n : \sum_i x_i = x} p^x (1-p)^{n-x} \\
&= \left| \left\{ (x_1, \ldots, x_n) \in \{0,1\}^n : \sum_{i=1}^n x_i = x \right\} \right| p^x (1-p)^{n-x}.
\end{aligned}$$

So it remains to compute the number of binary sequences $(x_1, \ldots, x_n)$ for such $\sum_{i=1}^n x_i = x$.[7] From Example 3.13, this number is the binomial coefficient $\binom{n}{x}$. Plugging this into the above expression gives (4.23). ☐

Definition 4.39 (Binomial distribution)**.** *Let $n \geq 1$ be an integer and $p \in [0, 1]$. A discrete random variable X is said to have the* Binomial(n, p) *distribution, written $X \sim$* Binomial(n, p)*, if its probability mass function is given by the right hand side of* (4.23)*.*

Thus Theorem 4.38 can be rephrased as follows:

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p) \Rightarrow X_1 + \cdots + X_n \sim \text{Binomial}(n, p). \tag{4.24}$$

The distribution of $X_1 + \cdots + X_n$ is called the *binomial* distribution because its probability mass function is given by the terms of a binomial expansion: By the binomial theorem (3.10), we have

$$1 = 1^n = ((1-p) + p)^n = \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x}.$$

This also verifies that the right hand side of (4.23) indeed defines a probability mass function, i.e., the sum of probabilities is 1.

We end this section by further extending Definition 4.32 to an *infinite* sequence of random variables.

Definition 4.40 (Independence and i.i.d. for an infinite sequence of random variables)**.** *Let $X_1, X_2, \ldots$ be a sequence of random variables. We say that they are jointly independent if $X_1, \ldots, X_n$ are jointly independent for all n. If in addition they have the same distribution, we say that they are independent and identically distributed.*

Remark 4.41 (Technical comment). Consider the infinite dimensional random vector $(X_1, X_2, \ldots)$. It is a mathematical fact (whose proof is beyond the scope of this course) that the joint distribution of $X_1, X_2, \ldots$ is uniquely determined by the joint distributions of the finite sequences $X_1, \ldots, X_n$ where n ranges over all positive integers. Intuitively, this is why in Definition 4.40 we only impose the joint independence of $X_1, \ldots, X_n$ for all n.

Remark 4.42. We may also consider *conditional independence of random variables* (given an event or more generally other random variables). An example is given in Exercise 7 in Section A.4.1.

[7] This is exactly the situation described in point (ii) in Section 3.3.